

Customer Shopping Behavior Analysis

1. Project Introduction

This project focuses on analyzing customer purchasing patterns using a transactional dataset containing 3,900 records across multiple product categories. The objective is to identify trends related to spending habits, customer segmentation, product popularity, and subscription usage to support informed business decisions.

2. Dataset Overview

- - Total Records: 3,900
- - Number of Columns: 18

Main Attributes Included

Customer Information

- - Age
- - Gender
- - Location
- - Subscription Status

Purchase Details

- - Item Purchased
- - Category
- - Purchase Amount
- - Season
- - Size
- - Color

Behavioral Variables

- - Discount Applied
- - Promo Code Used
- - Previous Purchases
- - Purchase Frequency
- - Review Rating
- - Shipping Type

Missing Values

- - The Review Rating column contained 37 null values.

3. Exploratory Data Analysis (Python)

The initial data preparation and exploration were performed using Python.

Data Import

The dataset was loaded into Python using the pandas library.

Preliminary Examination

- - `df.info()` was used to inspect data types and non-null counts.
- - `df.describe()` was used to generate summary statistics.

Handling Missing Data

Missing entries in the Review Rating column were filled using the median review rating for each product category.

Standardizing Column Names

Column names were converted to snake_case format to improve readability and consistency.

Feature Creation

Two new columns were engineered:

- - `age_group` – Created by grouping customer ages into categories.
- - `purchase_frequency_days` – Derived from purchase frequency information.

Redundancy Check

It was confirmed that `discount_applied` and `promo_code_used` conveyed the same information. As a result, `promo_code_used` was removed.

PostgreSQL Integration

The cleaned dataset was transferred from Python into PostgreSQL for further SQL-based analysis.

4. SQL Analysis in PostgreSQL

- Revenue by Gender

Female: 75,191

Male: 157,890

- High-Spending Customers Using Discounts

Total Customers: 839

- Top 5 Products by Average Rating

- - Gloves: 3.86
- - Sandals: 3.84
- - Boots: 3.82
- - Hat: 3.80
- - Skirt: 3.78

- Shipping Type Comparison

- - Standard: 58.46
- - Express: 60.48

- Subscribers vs Non-Subscribers

- - Yes: 1,053 customers | Avg Spend: 59.49 | Total Revenue: 62,645.00
- - No: 2,847 customers | Avg Spend: 59.87 | Total Revenue: 170,436.00

- Products with Highest Discount Usage

- - Hat: 50.00%
- - Sneakers: 49.66%
- - Coat: 49.07%
- - Sweater: 48.17%
- - Pants: 47.37%

- Customer Segmentation

- - Loyal: 3,116
- - New: 83
- - Returning: 701

- Top 3 Best-Selling Products in Each Category

Accessories: Jewelry (171), Sunglasses (161), Belt (161)

Clothing: Blouse (171), Pants (171), Shirt (169)

Footwear: Sandals (160), Shoes (150), Sneakers (145)

Outerwear: Jacket (163), Coat (161)

- Repeat Buyers and Subscription Adoption

- - No: 2,518
- - Yes: 958

- Revenue Contribution by Age Group

- - Young Adult: 62,143
- - Middle-aged: 59,197
- - Adult: 55,978
- - Senior: 55,763

5. Power BI Dashboard

An interactive dashboard was created in Power BI to visually communicate the analysis results. It includes:

- - 3.9K total customers
- - \$59.76 average purchase amount
- - 3.75 average review rating

Additional visuals display:

- - Revenue by category
- - Sales by category
- - Customer distribution by subscription status
- - Revenue by age group
- - Sales by age group

6. Strategic Recommendations

- - Increase Subscription Adoption: Promote subscriber-only perks and benefits.
- - Strengthen Customer Loyalty: Introduce reward programs for repeat buyers.
- - Reassess Discount Strategy: Monitor discount effectiveness to maintain profitability.
- - Promote High-Performing Products: Highlight highly rated and top-selling items.
- - Target High-Value Segments: Focus on high-revenue age groups and express-shipping customers.